CHANAKYA FELLOWSHIP

Crop Research Documentation Report

Methodology, Experiments, Results and Reproducibility Record

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This document is maintained as a permanent artifact record for TIH IIT Bombay. It should allow any future researcher to understand what was done, on which crops, with which models, and how to reproduce or benchmark against the reported results.

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1. INTRODUCTION

The fellowship work focused on applying AI/ML and computer vision techniques to crop disease identification using image-based datasets. The work covered three crops: tomato, mango, and soybean. A major part of the experimentation involved Vision-Language Models (VLMs), with Qwen2.5-VL-7B-Instruct used for crop-specific model training and evaluation. Earlier experimentation also included multiple multimodal architectures as part of the model-development process.

For tomato, the work included a dedicated company-dataset classification pipeline using Qwen2.5-VL-7B-Instruct with QLoRA fine-tuning, controlled train-validation-test splitting, validation monitoring, and final held-out testing. The final company-only tomato workflow was kept separate from the broader research pipeline and was designed specifically for multi-class classification.

For mango, model training and evaluation were carried out as part of the crop disease classification work. For soybean, the work consisted of training the model on the VDI infrastructure using the dataset and training pipeline that were provided for the task.

Overall, the fellowship involved practical work in model training, evaluation, dataset handling, and reproducible experimentation across multiple crop datasets.

1.1 Scope of Work

This report covers the crop-related AI/ML work carried out for tomato, mango, and soybean, including model training, evaluation, experiment results, and reproducibility details where available. The tomato and mango sections document the respective disease-classification experiments and model-development work. For soybean, the scope is limited to training the model on the VDI using the dataset and training pipeline provided for the task. The report does not claim development of the underlying soybean training pipeline and does not cover work outside these three crop-specific tasks.

2. CROPS WORKED ON

#	Crop Name	Task / Focus Area	Status
1	Tomato	Disease and pest classification using Vision-Language Models	Completed
2	Mango	Leaf disease classification using Vision-Language Models	Completed

3. DETAILED METHODOLOGY

3.1 Crop: Tomato

Models Used

The following models and architectures were investigated during the tomato work:

- LLaVA 1.5-7B
- LLaVA 1.6 Mistral-7B
- BLIP-2 OPT-2.7B
- Florence-2-base-ft
- Qwen2.5-VL-7B-Instruct - fine-tuned
- Qwen2.5-VL-7B-Instruct (4-bit) - zero-shot benchmark
- Qwen3-VL-8B-Instruct (4-bit)

- Gemma-3-12B-it (4-bit)
- InternVL3-8B (4-bit)
- LLaMA-3.2-11B-Vision (4-bit)
- Molmo-7B-D-0924 (4-bit)
- DeepSeek-VL2-Small (4-bit)
- CLIP — out-of-distribution screening
- Qwen2.5-VL-7B-Instruct with 4-bit QLoRA - TIH-provided tomato dataset

The project progressed through multiple VLM architectures before Qwen2.5-VL-7B-Instruct was selected as the main fine-tuned model. A separate seven-model zero-shot benchmark and CLIP-based OOD screening stage were also evaluated.

Methodology

The initial tomato experiments used the PlantVillage tomato subset containing 10 disease and healthy classes. Since the dataset provided images and class labels but no textual descriptions, structured diagnostic captions were generated using multiple templates per disease class. The dataset was shuffled using random seed 42 and divided using a 90:10 training-validation split, with augmentation applied only to the training data.

Multiple VLM architectures were evaluated sequentially. LLaVA 1.5-7B and LLaVA 1.6 Mistral-7B encountered batch-collation incompatibilities, BLIP-2 OPT-2.7B trained successfully but did not reliably maintain the required structured output, and Florence-2-base-ft was discontinued because of environment compatibility issues. Qwen2.5-VL-7B-Instruct was selected as the final architecture and trained for 3 epochs using LoRA rank 32, alpha 64, batch size 4, gradient accumulation 4, learning rate 2×10^{-4}, AdamW, and cosine scheduling. Evaluation included classification, caption-generation, and severity-estimation metrics.

A separate zero-shot benchmark was conducted using seven VLMs on 30 tomato images across six disease classes, using a fixed structured prompt and expert-validated reference captions. CLIP-based out-of-distribution screening was also incorporated into the diagnostic pipeline.

A later experiment used a tomato dataset provided by TIH for direct 11-class classification. The dataset contained 7,095 raw images, with 6,545 usable images after cleanup and deduplication. These were divided into 5,247 training, 649 validation, and 649 test images using a deterministic, stratified split. Qwen2.5-VL-7B-Instruct was fine-tuned using 4-bit QLoRA, with a maximum of 3 epochs, learning rate 5×10^{-5}, micro-batch size 1, and gradient accumulation of 8. The model was instructed to return only the predicted class label, and final evaluation was performed on the held-out test set.

Experiment Versions

Version	Date	Change Description	Outcome
v1.0	5 June 2026	LLaVA 1.5-7B evaluated for structured tomato diagnosis	Abandoned due to batch-collation incompatibility
v2.0	8 June 2026	LLaVA 1.6 Mistral-7B evaluated	Abandoned; same incompatibility reproduced
v3.0	11 June 2026	BLIP-2 OPT-2.7B trained as baseline	Training converged; structured outputs were unreliable
v4.0	20 June 2026	Florence-2-base-ft evaluated as an alternative VLM	Abandoned due to repeated environment incompatibilities
v5.0	22 June 2026	Qwen2.5-VL-7B-Instruct fine-tuned for structured tomato diagnosis	Selected as the best-performing configuration

v6.0	25 June 2026	Zero-shot benchmark using seven VLMs	Benchmark conducted; four models evaluated against expert references
v7.0	9 July 2026	CLIP-based out-of-distribution screening	OOD screening integrated; threshold calibrated at 0.22
v8.0	3 August 2026	Qwen2.5-VL-7B-Instruct fine-tuned using 4-bit QLoRA on the TIH-provided 11-class tomato dataset	87.98% test accuracy and 0.8991 Macro-F1

Results & Metrics

Model / Configuration	Accuracy	Precision	Recall	F1-score
LLaVA 1.5-7B	N/A	N/A	N/A	N/A
LLaVA 1.6 Mistral-7B	N/A	N/A	N/A	N/A
BLIP-2 OPT-2.7B	N/A	N/A	N/A	N/A
Florence-2-base-ft	N/A	N/A	N/A	N/A
Qwen2.5-VL-7B-Instruct - PlantVillage	98.50%	0.9836	0.9823	0.9827 Macro-F1
Qwen2.5-VL-7B-Instruct + 4-bit QLoRA - TIH dataset	87.98%	0.8930 Macro	0.9186 Macro	0.8991 Macro-F1

For BLIP-2, classification metrics were not reported because the generated responses did not consistently preserve the required structured format. It achieved BLEU-1 = 0.2081, BLEU-4 = 0.0819, and ROUGE-L = 0.3017.

The PlantVillage Qwen2.5-VL experiment also achieved:

Metric	Value
BLEU-1	0.6047
BLEU-4	0.4556
ROUGE-L	0.5869
CLIP Text Cosine	0.3136
Severity R^2	0.9654

Severity MAE	0.025

Six of the ten tomato classes obtained **F1 = 1.0000**, while Target Spot had the lowest class-level F1 at **0.8966**.

For the TIH-provided dataset experiment:

- Weighted F1: **0.8807**
- Unparseable outputs: **0**
- Average inference latency: **~0.29 s/image**

Reproducibility - Dataset, Model & Split Artifacts

Dataset Path	TIH-provided tomato dataset
Train/Val/Test Split File	Not included
Split Method & Seed	80/10/10 stratified by class, random_seed=20260911
Model Checkpoint Path	checkpoints/tih_tomato_qwen_qlora_strong
Code Repository / Commit	https://github.com/iAakash1/experimentation - Branch: feat/tih-tomato-qwen, Commit: 8eb2cd6
Environment / Config File	configs/train/qwen25vl_tih_tomato_strong.yaml

Key Observations / Learnings

- LLaVA 1.5 and LLaVA 1.6 reproduced the same batch-collation incompatibility, showing that the issue was architectural rather than specific to one version.
- BLIP-2 successfully converged but could not consistently generate the required structured diagnostic format.
- Florence-2 was abandoned after repeated environment and compatibility failures.
- Qwen2.5-VL-7B-Instruct achieved the strongest result on the PlantVillage evaluation, with 98.50% accuracy and 0.9827 Macro-F1 while also supporting structured diagnostic generation.
- The zero-shot benchmark revealed that successful inference does not necessarily mean valid output: Qwen3-VL-8B generated degenerate repeated-character outputs on 16 of 30 images.
- The TIH-provided dataset experiment demonstrated the use of Qwen2.5-VL for a simpler direct classification task, achieving 87.98% held-out test accuracy and 0.8991 Macro-F1 with zero unparseable outputs. Its main error pattern was confusion of Late Blight with Leaf Curl, Leaf Miner, and Target Leaf Spot.

3.2 Crop: Mango

Models Used

- Qwen2.5-VL-7B
- Qwen2.5-VL-7B with QLoRA fine-tuning

The same Qwen2.5-VL-7B backbone and fine-tuning approach used in the comparative crop evaluation was applied separately to the mango dataset.

Methodology

The mango experiment used the MangoLeafBD dataset, containing approximately 4,000 mango leaf images across 8 classes. A knowledge-grounded instruction dataset was generated using a curated Disease Knowledge Base, controlled vocabulary, concept modelling, and template-based caption generation rather than externally generated captions.

Qwen2.5-VL-7B was fine-tuned using QLoRA with the MLX framework on Apple Silicon hardware with 24 GB unified memory. The LoRA rank was 16. A fixed, image-grouped held-out test split was used so that images from the same leaf did not appear across training and testing data.

Evaluation was performed on a held-out test set of 200 images. Disease classification was evaluated using accuracy, Macro-F1, and Weighted-F1. Generated diagnostic captions were additionally evaluated using BLEU-4, ROUGE-L, METEOR, CIDEr, and BERTScore-F1.

Experiment Versions

Version	Date	Change Description	Outcome
Mango Baseline	23 July 2026	Qwen2.5-VL-7B evaluated without crop-specific fine-tuning	17.5% test accuracy
Mango QLoRA	29 July 2026	Qwen2.5-VL-7B fine-tuned on MangoLeafBD using the knowledge-grounded instruction dataset	82.0% test accuracy, Macro-F1 0.811

Results & Metrics

Model / Configuration	Accuracy	Precision	Recall	F1-score
Qwen2.5-VL-7B - Base	17.5%	Not reported	Not reported	Not reported
Qwen2.5-VL-7B + QLoRA - MangoLeafBD	82.0%	Not reported	Not reported	0.811 Macro-F1

The fine-tuned model also achieved a Weighted-F1 of 0.811, representing an absolute accuracy improvement of 64.5 percentage points over the base model.

For caption generation, the fine-tuned mango model achieved:

Metric	Base	Fine-tuned
BLEU-4	0.004	0.190
ROUGE-L	0.098	0.423
METEOR	0.160	0.454
CIDEr	0.000	0.946
BERTScore-F1	0.843	0.907

Reproducibility — Dataset, Model & Split Artifacts

Dataset Path	MangoLeafBD
Train/Val/Test Split File	Not included
Split Method & Seed	Frozen image-grouped held-out test split
Model Checkpoint Path	checkpoints/qwen25vl_mango_qlora
Code Repository / Commit	To be added after repository verification
Environment / Config File	MLX framework, Apple Silicon, 24 GB unified memory

Key Observations / Learnings

- Fine-tuning improved mango disease classification accuracy from 17.5% to 82.0%.
- Both Macro-F1 and Weighted-F1 were 0.811, indicating reasonably consistent performance across the eight mango classes.
- Caption-generation metrics improved across all five reported measures after fine-tuning.
- The report notes that the mango dataset was smaller and contained classes with more visually similar disease characteristics, making classification more challenging than the tomato dataset.
- The evaluation used a controlled dataset; performance under more varied field conditions was not established in this experiment.

4. CONCLUSION

The work across tomato and mango showed that Qwen2.5-VL-based fine-tuning was effective for crop disease diagnosis across different datasets and class structures. For tomato, the structured-diagnosis experiment achieved 98.50% accuracy and 0.9827 Macro-F1, while the later experiment on the TIH-provided 11-class tomato dataset achieved 87.98% accuracy and

0.8991 Macro-F1. For mango, Qwen2.5-VL fine-tuned using QLoRA achieved 82.0% accuracy and 0.811 Macro-F1, compared with 17.5% accuracy from the base model.

Across both crops, Qwen2.5-VL with parameter-efficient fine-tuning gave the most consistent results. The experiments also showed that crop-specific training data and structured supervision can improve both disease classification and diagnostic text generation.

The main limitations were the variation in performance across datasets and the difficulty of distinguishing classes with visually similar symptoms. In the TIH tomato experiment, Late Blight was commonly confused with Leaf Curl, Leaf Miner, and Target Leaf Spot. For mango, the smaller dataset and similarity between disease classes made classification comparatively more challenging.

4.1 Recommendations for Future Work

- Extend the Qwen2.5-VL fine-tuning approach to additional crops using the same crop-specific training pipeline.
- Increase the number of annotated samples for disease classes that show higher confusion, particularly visually similar classes.
- Improve class balancing and targeted data augmentation for underrepresented or difficult disease categories.
- Benchmark the same datasets using newer Vision-Language Models and compare their accuracy, F1-score, training efficiency, and inference performance.
- Further refine the structured diagnostic output so that disease classification and generated explanations remain consistent across different crops.

4.2 Acknowledgements

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Appendix A. CONSOLIDATED ARTIFACT INDEX

Crop	Dataset Path	Split Path	Model Path	Repo / Commit
Tomato	TIH-provided tomato dataset	Not included	checkpoints/tih_tomato_qwen_qlora_strong	https://github.com/iAakash1/experimentation - Branch: feat/tih-tomato-qwen, Commit: 8eb2cd6
Mango	MangoLeafBD	Not included	checkpoints/qwen25vl_mango_qlora	https://github.com/iAakash1/experimentation